Linux Commands Practical Guide

Step-by-step hands-on guide for Ubuntu in a virtual machine

Compiled from the Linux practical steps discussed in this conversation and organized for practice. Includes commands, explanations, expected results, and safety notes.

Environment: Ubuntu Linux running inside a virtual machine

Recommended workspace: `~/mlops/linux_practical`

> Run one command at a time. Paths and outputs can vary. Do not use sudo to bypass permission problems unnecessarily.

Before you begin: Set up your workspace

Open Ubuntu and press Ctrl + Alt + T to open Terminal. The tilde (~) represents your home directory.

Permission issue: /usr/local/mlops

If creating files in /usr/local/mlops gives “Permission denied”, practise in your home directory instead.

```
cd ~
mkdir -p ~/mlops/linux_practical
cd ~/mlops/linux_practical
pwd
```

Expected path is usually /home/vp/mlops/linux_practical if your username is vp.

Check your environment

```
whoami
pwd
ls -la
```

whoami prints your username; pwd prints the current directory; ls -la lists visible and hidden files with details.

Practical 1 — File and directory navigation

pwd — current directory

```
pwd
```

ls — list files

```
ls
ls -l
ls -la
ls -lh
ls -lt
```

-l shows details, -a includes hidden files, -h uses human-readable sizes, and -t sorts by modification time.

cd — change directory

```
cd ~
cd ~/mlops/linux_practical
cd ..
cd -
```

cd .. moves up one directory; cd - returns to the previous directory.

find — search for files

```
find . -name "*.txt"
find . -type d
find . -type f -size +1k
```

The dot (.) means search from the current directory. The final command lists files larger than 1 KB.

which / whereis — locate a command

```
which python3
which pip
whereis python3
```

Practical 2 — Create, copy, rename and remove files

Create directories and files

```
mkdir models
mkdir data
touch requirements.txt
ls -l
```

If a directory already exists, use mkdir -p models or continue if it is already there.

Write and view a file

```
echo "Linux practical practice" > requirements.txt
```

```
cat requirements.txt
```

> writes output to a file and overwrites existing contents.

Copy files

```
echo "Sample notes" > notes.txt
cp notes.txt notes_backup.txt
ls
```

cp copies a file. Use cp -r source_dir destination_dir to copy a directory.

Rename or move files

```
mv notes_backup.txt notes_copy.txt
mkdir -p reports
mv notes_copy.txt reports/
ls reports
```

mv renames a file or moves it to another location.

Remove a file

```
ls notes.txt
rm notes.txt
```

rm permanently deletes files. Check the exact path with ls before removing anything. Avoid rm -rf unless certain of the target.

Nested directories

```
mkdir -p experiments/run_01/checkpoints
find experiments -type d
```

Practical 3 — View, edit and search file contents

Create a sample log

```
cat > training.log <<'EOF'
INFO: Training started
INFO: Loading dataset
INFO: Model initialized
ERROR: Missing values detected
INFO: Data cleaning completed
ERROR: Validation failed
INFO: Training finished
EOF
```

This creates a seven-line log file. The EOF marker ends the input block.

View contents

```
cat training.log
head -n 3 training.log
tail -n 2 training.log
less training.log
```

In less, use arrow keys to move and press q to quit.

Search text

```
grep "ERROR" training.log
grep -n "ERROR" training.log
grep -i "error" training.log
```

grep searches matching lines; -n includes line numbers and -i ignores case.

Edit using nano

```
nano notes.txt
```

Type a line. Press Ctrl+O to save, Enter to confirm, and Ctrl+X to exit. Verify with:

```
cat notes.txt
```

Count lines

```
wc -l training.log
```

Expected count: 7 lines.

Practical 4 — Permissions and shell scripts

Inspect and change permissions

```
ls -l notes.txt
chmod 644 notes.txt
ls -l notes.txt
```

Mode 644 usually appears as -rw-r--r--: owner can read/write; group and others can read.

Make a script executable

```
echo 'echo "Hello from Linux"' > hello.sh
chmod +x hello.sh
./hello.sh
```

Expected output: Hello from Linux. The +x adds execute permission.

sudo and chown

```
sudo apt update
# Example only: replace user/group and target with correct values
sudo chown user:user file.txt
```

Use sudo only when administrator privileges are required. Do not run the placeholder chown command literally or change ownership of system directories to work around a permission error.

Practical 5 — Process and job management

View processes

```
ps
ps aux
ps aux | grep python
```

ps aux lists processes system-wide. grep may appear in its own results.

Live process monitor

```
top
```

Press q to exit.

Background job, jobs, fg and kill

```
sleep 300 &
jobs
kill %1
jobs
```

& starts a background job. jobs lists jobs in this shell. kill %1 requests termination of job 1; use the number shown by jobs.

nohup with log output

```
nohup bash -c 'for i in 1 2 3 4 5; do echo "Run $i"; sleep 2; done' > nohup.log 2>&1 &
cat nohup.log
```

The log may be incomplete if checked immediately; wait a few seconds and run cat again.

Practical 6 — System and storage monitoring

Disk space and directory size

```
df -h
du -sh ~/mlops/linux_practical
du -h --max-depth=1 . | sort -h
```

df -h shows filesystem space. du summarizes directory usage. The last command sorts immediate subdirectory sizes.

RAM

```
free -h
```

Displays memory and swap usage.

GPU status, if supported

```
nvidia-smi
watch -n 1 nvidia-smi
```

If no NVIDIA GPU or driver is available in the VM, nvidia-smi may fail. Press Ctrl+C to stop watch.

Practical 7 — Archive and compression

Create sample data

```
mkdir -p dataset
echo "Sample dataset for MLOps" > dataset/data.txt
```

Create and inspect tar.gz

```
tar -czvf dataset.tar.gz dataset/
ls -lh dataset.tar.gz
```

Options: c=create, z=gzip compression, v=verbose, f=archive filename.

Extract the archive

```
mkdir -p extracted
tar -xzvf dataset.tar.gz -C extracted
cat extracted/dataset/data.txt
```

x extracts the archive; -C selects the destination directory.

ZIP and unzip

```
zip -r project.zip dataset/
unzip project.zip -d zip_extracted
ls zip_extracted/dataset
```

If zip/unzip are missing: sudo apt update then sudo apt install zip unzip.

Practical 8 — Pipes and text processing

Pipe output into another command

```
cat training.log | grep "ERROR" | tail -n 20
```

The pipe symbol | sends one command's output to the next.

Redirection

```
echo "Practical completed" > result.txt
echo "Linux commands practiced" >> result.txt
cat result.txt
```

> overwrites a file; >> appends to it.

Sort and count repeated lines

```
printf "apple\nbanana\napple\norange\nbanana\napple\n" > fruits.txt
sort fruits.txt | uniq -c
```

uniq counts adjacent repeated lines, so it is commonly used after sort.

awk and sed

```
printf '101,Asha\n102,Ravi\n' > data.csv
awk -F',' '{print $2}' data.csv
sed 's/Ravi/Rohan/g' data.csv
```

awk prints a selected field; sed performs a text substitution. This sed command prints changed text without modifying the file.

Practical 9 — Networking and remote access

Connectivity and web requests

```
ping -c 4 8.8.8.8
curl -I https://example.com
wget https://example.com
head index.html
```

ping sends four requests. curl -I requests headers. wget downloads a page, commonly saved as index.html. Network restrictions can affect results.

Network interfaces

```
ip addr
```

SSH and SCP syntax

```
ssh user@server-address
scp model.pt user@server-address:/home/user/models/
scp -r ./dataset user@server-address:/data/
```

SSH and SCP require a real remote host and valid credentials or SSH keys. Replace placeholders with details supplied by your instructor.

If curl or wget is missing, install it with sudo apt update and sudo apt install curl wget.

Practical 10 — Git basics

Check and configure Git

```
git --version
git config --global user.name "Your Name"
git config --global user.email "your-email@example.com"
git config --global --list
```

Replace the sample name and email with your own details.

Initialize a repository

```
cd ~/mlops/linux_practical
git init
git status
```

Stage and commit

```
echo "My first Git practical" > git_notes.txt
git add git_notes.txt
git commit -m "Add Git practical notes"
git log --oneline
```

Inspect changes

```
echo "Practising Git commands" >> git_notes.txt
git diff
git status
git add git_notes.txt
git commit -m "Update Git practical notes"
```

Create and switch branches

```
git branch
git checkout -b feature/practical
git branch
```

Switch back using the branch name shown by git branch, often main or master: git checkout main.

Remote repository commands

```
git clone https://github.com/OWNER/REPOSITORY.git
git pull origin main
git push origin main
```

Remote commands require a real repository URL and access. Push may require authentication and a configured remote.

Practical 11 — MLOps-specific command patterns

Background job with logging

```
mkdir -p logs
nohup python train.py --epochs 50 > logs/train.log 2>&1 &
```

This syntax requires train.py to exist, dependencies to be installed, and the script to accept --epochs. Use the harmless loop demo in Practical 5 for safe practice.

Watch system metrics

```
watch -n 1 free -h
watch -n 1 nvidia-smi
```

Run one monitor at a time and press Ctrl+C to exit.

Persistent sessions with tmux

```
tmux new -s training
# Detach: press Ctrl+B, then D
tmux attach -t training
```

Install tmux if needed with sudo apt install tmux.

Find large checkpoints

```
find ./checkpoints -name "*.ckpt" -printf '%s %p\n' | sort -nr | head -10
```

Lists up to ten largest matching files. The directory must exist.

Check a service port

```
sudo lsof -i :8000
curl http://localhost:8000/health
```

These check a process using port 8000 and query a health endpoint, if a service is running there.

Safe local server demo

```
python3 -m http.server 8000
```

This runs a simple web server in the foreground. In a second terminal, run curl http://localhost:8000. Stop the server with Ctrl+C in the first terminal.

Quick reference

Command	Purpose
pwd / ls / cd	Show location, list items, change directory
mkdir / touch	Create directories / empty files
cp / mv / rm	Copy / move or rename / delete
cat / head / tail / less	View file contents
grep / find / wc	Search text / find paths / count lines, words, characters
chmod / chown / sudo	Permissions / ownership / administrator command
ps / top / kill / jobs	Inspect and manage processes and jobs
df / du / free / watch	Monitor disk, directory sizes, memory, repeat commands
tar / zip / unzip	Archive, compress and extract files
ping / curl / wget / ssh / scp	Network checks, web requests, downloads, remote login/copy
git status / add / commit / log / branch	Inspect and manage version history
| / > / >>	Pipe output / overwrite file / append to file

Final verification checklist

- ■ I can navigate using pwd, ls and cd.
- ■ I can create, copy, rename, move and remove test files.
- ■ I can view/search files using cat, head, tail and grep.
- ■ I can change permissions with chmod and run a shell script.
- ■ I can inspect processes and manage background jobs.
- ■ I can check disk and memory with df, du and free.

- ■ I can create and extract tar.gz and zip archives.
- ■ I can use pipes, redirection, sort, uniq, wc, awk and sed.
- ■ I understand basic network checks and SSH/SCP syntax.
- ■ I can initialize a Git repository, commit changes and create a branch.
- ■ I understand that remote, GPU, port and training commands need the relevant environment.

Practice safely in your home-directory workspace. Avoid deleting unfamiliar files, changing ownership of system folders, or running destructive commands with sudo. Some commands depend on installed packages, hardware, network access, or a configured remote repository.